

Game Physics

$$v(t) = \frac{dx(t)}{dt}, \quad a(t) = \frac{dv(t)}{dt}$$

$F=ma$ - Newton's Second Law

Гравитация: $F = (0, -mg)^T$

Уравнение для гравитации:

$$x(t) = \frac{1}{2}gt^2 + v_0t + x_0$$

- на месте g может быть любое податное ускорение

$$v(t) = \int a dt = at + v_0$$

↖ a это константа, не функция, т.к. гравитация не меняется

$$\begin{aligned} x(t) &= \int v(t) dt = \int (at + v_0) dt = \int at dt + \int v_0 dt = a \int t dt + v_0 t + x_0 = \\ &= \frac{at^2}{2} + v_0 t + x_0. \end{aligned}$$

Mass Spring Systems

System dynamics:

Осциллы, веревки, эластичные объекты

Внутренние силы F_{int} , действующие м/г точками, и внешние силы, действующие на точки, определяют суммарную силу для каждой точки:

$$F_i = \sum_j F_{int,ij} + F_{ext,i}$$

↑ м/г точки i и j

Uniform Mass Distribution:

Если объект имеет массу M и n точек, то:

$$m = \frac{M}{n} \quad \text{— где каждая точка}$$

Hooke's Law (^{упругие} Elastic Forces):

Force needed to extend (or compress) a spring by a distance is proportional to that distance



$$F = -k(l - L) \quad \text{— с такой силой пружина растягивает (если } l > L \text{), если сжимает}$$

k — жесткость, l — изменение длины пружины

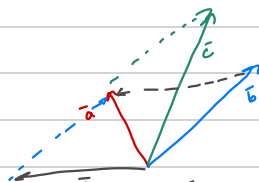
k — stiffness (жесткость)

L — длина пружины в покое

l — текущая длина пружины

Сила пропорциональна изменению длины пружины и действует так, что пружина стремится к положению покоя.

Векторная форма:



Направленная сила.

$$F_{ij} = -k(l - L) \frac{x_i - x_j}{l}$$

$d_{ij} = |x_i - x_j|$ — длина пруж. (точка)
 l — длина пруж.
 $l = \|x_i - x_j\|$ — длина пруж.
 j действует на i

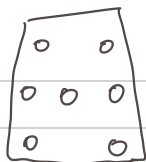
$l = \|x_i - x_j\|$
 вектор делит на его длину, получаем единичн. вектор, тогда у F было направление

F_{ij} — сила, с которой действует пружина на точки i и j .

$$F_{ij} = -F_{ji}$$

$$F_i = -F_{ij}$$

$$F_j = F_{ij}$$



$$m_i = \frac{M}{n}$$

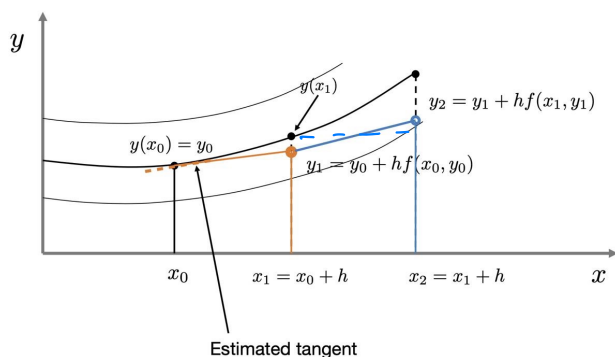
$$\sum_j F_{ij}^{\text{int}} + F_i^{\text{ext}} = m a_i$$

Euler's Method:

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$y_0 + h \cdot y_0$$

$$y_0 + h \cdot y_0 + h \cdot f(y, x)$$



Heun's Method:

Do a step with an averaged tangent

- based on current step

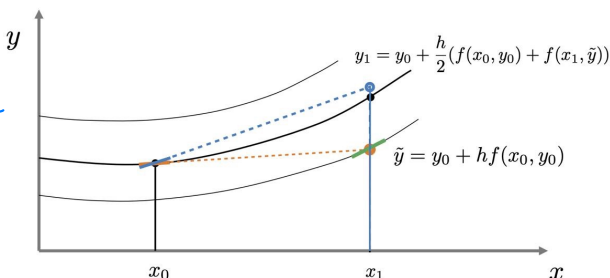
- and from an Euler step

Second order

$$\tilde{y} = y_0 + h f(x_0, y_0)$$

$$y_1 = y_0 + \frac{h}{2} (f(x_0, y_0) + f(x_1, \tilde{y}))$$

no cycle yet and a
even average mit gespro
vorher



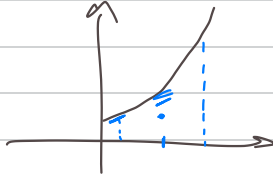
Predictor-corrector method:

predict slope, then correct &
step forward

Midpoint Method

$$\tilde{y} = y_0 + \frac{h}{2} f(x_0, y_0)$$

$$y_1 = y_0 + h f(x_0 + \frac{h}{2}, \tilde{y})$$



Runge-Kutta Methods

Family of methods, we use RK4

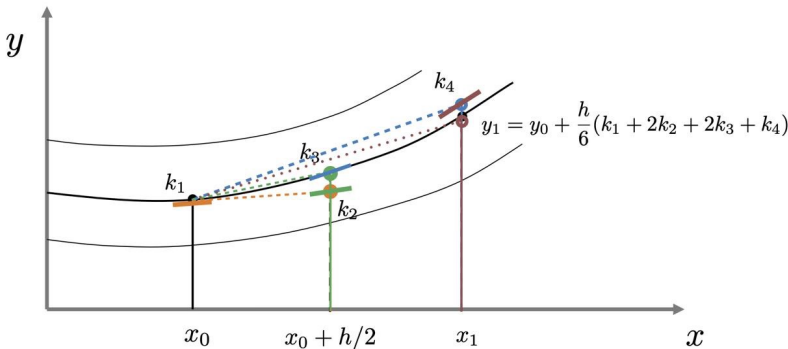
$$k_1 = f(x_0, y_0)$$

$$k_2 = f(x_0 + \frac{h}{2}, y_0 + \frac{h}{2} k_1) - \text{predict midpoint}$$

$$k_3 = f(x_0 + \frac{h}{2}, y_0 + \frac{h}{2} k_2) - \text{correct the estimate}$$

$$k_4 = f(x_0 + h, y_0 + h k_3) - \text{predict with full step}$$

$$y_1 = y_0 + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4) - \text{move step with weighted slopes}$$



Leap-Frog Method:

Significantly more stable than explicit Euler.

$$v(t + \frac{h}{2}) = v(t - \frac{h}{2}) + h \cdot a(t)$$

$$x(t + h) = x(t) + h \cdot v(t + \frac{h}{2})$$

Rigid Bodies

Center of mass:

$$x_{cm} = \frac{\sum_i m_i x_i}{\sum_i m_i}$$

center of mass

Центр масс негет себе сам, дуга
век объект. Масса и зроще нере
масса парна масса нере объекта

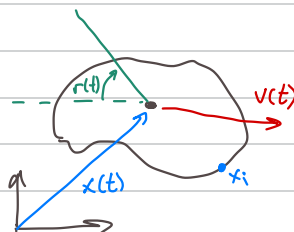
$$M = \sum_i m_i$$

$$v_{cm} = \frac{\sum_i m_i v_i}{M}$$

$$a_{cm} = \frac{\sum_i m_i a_i}{M}$$

absolut position

$$x_i^{world} = x_{cm}(t) + Rot_{rel}(x_i)$$



World space

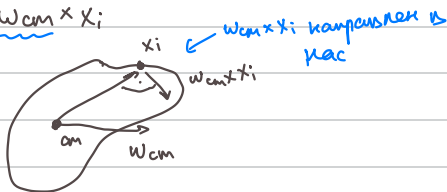
Body space



Total velocity:

$$v_i = v_{cm} + \omega_{cm} \times x_i$$

зроще гомеи
крупноте по
кратчайшему
пути нроче
качелон



$$u \times v = \begin{pmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{pmatrix}$$

$$\begin{pmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix} \begin{pmatrix} 2 & 3 & -3 & 2 \\ 3 & 1 & -1 & 3 \\ 1 & 2 & -2 & 1 \end{pmatrix}$$

DIO TOO

В 2D пространство вращения вокруг Z, поэтому ω - число, обычно
▷ радиан/сек

Velocity of a point:

$$\mathbf{v}_i = \begin{pmatrix} -\omega x_{i,y} \\ \omega x_{i,x} \end{pmatrix}$$

$\mathbf{p} = m\mathbf{v}$ - импульс, показывает, как сильно изменить движение тела

$$\text{Rot}_\alpha = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$$

Euler step for 2D:

$$\mathbf{x}_{cm} = \mathbf{x}_{cm} + h \mathbf{v}_{cm}$$

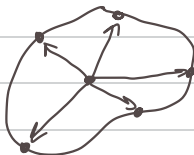
$$\mathbf{v}_{cm} = \mathbf{v}_{cm} + h \boldsymbol{\omega}_{cm}$$

Inertia Tensor in 2D:

$$I = \sum_n m_n x_n \cdot x_n$$

↑
радиус-вектор до x_n от CM

Всегда считать relative to cm координатами!!!



Момент инерции - сопротивление тела изменению его угловой скорости (чем дальше масса от оси, тем сильнее)

Torque:

Момент силы - способность силы вызывать вращение (чем дальше $\mathbf{r}$ - радиус-вектор силы, тем сильнее вращение)

of a mass point:

$$\mathbf{q}_i = \mathbf{x}_i \times \mathbf{f}_i$$

↑
радиус-вектор

↑
угловое ускор.

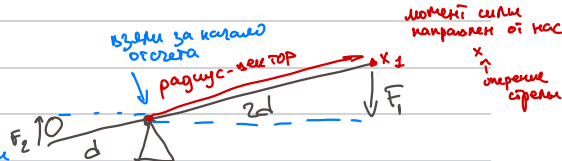
of a body:

$$\mathbf{q} = \sum_i \mathbf{q}_i$$

$$\mathbf{q} = I \boldsymbol{\alpha}$$

$$\text{аналог } F = ma$$

$$I - \text{аналог массы}$$



Angular momentum:

$$L = \sum_i \mathbf{x}_i \times (m_i \mathbf{v}_i) = I \boldsymbol{\omega}$$

↑
 $\mathbf{p}_i$ импульс точки x_i



$$-F_1 \cdot d_1 + F_2 \cdot d_2 = 0$$

$$F_2 \cdot d_1 = F_1 \cdot d_2$$

$$F_2 = 2F_1$$

learn

Если тело только вращ., но не движ., то $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$

Второй закон Ньютона для углового движения:

$$\frac{dL}{dt} = \tau$$

$$L^{(t+h)} = L^{(t)} + h\tau \quad \text{learn}$$

Изменение угловой скорости со временем:

$$\omega(t+h) = I^{-1} L(t+h)$$

Inertia vector is constant over time & just a number $\Rightarrow$

$$\omega(t+h) = \omega(t) + \frac{h\tau}{I} \quad \text{learn}$$

Collisions

Иногда смотрят перпендикуляр поверхности первого объекта collisions.
(но в лекциях - B)

Мгновенная скорость в точке столкновения i тела A:

$$\tilde{v}_{i,a} = v_a + \omega_a \times x_{i,a}$$

Считаем относительную скорость

$$v_{rel} = \tilde{v}_{i,a} - \tilde{v}_{j,b}$$

Считаем скалярное произведение относ. скорости (показано на нормаль):

$$v_{rel} = (\tilde{v}_{i,a} - \tilde{v}_{j,b}) \cdot n$$

Голкинг: $v_{rel} < 0$

Separating: $v_{rel} > 0$

Sliding contact: $v_{rel} = 0$

Impulse:

$$J = m \Delta v \quad (\text{no timestamp to avoid singularity})$$



Use c - coefficient of restitution ("коэффициент упругости")

$$v' \cdot n = -c v \cdot n$$

$$m_a v_a + m_b v_b = m_a v'_a + m_b v'_b \quad \text{закон сохранения импульса}$$

$$J_{lin} = \frac{-(1+c) v_{rel} \cdot n}{\frac{1}{m_a} + \frac{1}{m_b}}$$

v_{rel} - pre collision

v'_{rel} - post collision

$$v'_a = v_a + J_{lin} \frac{n}{m_a}$$

$$v'_b = v_b - J_{lin} \frac{n}{m_b}$$

$$J_{lin} = \frac{-(1+c) v_{rel} \cdot n}{\frac{1}{m_a} + \frac{1}{m_b}} \quad \text{— формула не работает, если относительная скорость равна 0}$$

Due to torque:

$$J_n = M \Delta \omega_{cm}$$

$$x_{cd} \times J_n = \Delta L = I \Delta \omega$$

Вращение из:

$$v'_{rel} \cdot n = -c v_{rel} \cdot n$$

$$(v'_a - v'_b) \cdot n = -c v_{rel} \cdot n$$

$$(v_a + J_{lin} \frac{n}{m_a} - v_b - J_{lin} \frac{n}{m_b}) \cdot n = -c v_{rel} \cdot n$$

$$v_a - v_b = v_{rel}$$

$$v_{rel} \cdot n + J_{lin} (\frac{1}{m_a} - \frac{1}{m_b}) n \cdot n = -c v_{rel} \cdot n$$

$$J_{lin} (\frac{1}{m_a} - \frac{1}{m_b}) n \cdot n = -c v_{rel} \cdot n - v_{rel} \cdot n$$

т.к. n - вектор нормали, $g_{nn} = 1$

Angular velocity update

$$\omega'_a = \omega_a + \frac{(x_a \times J_n)}{I_a}$$

$$\omega'_b = \omega_b + \frac{(x_b \times J_n)}{I_b}$$

learn

Total Impulse:

$$J = \frac{-(1+c) v_{rel} \cdot n}{\frac{1}{m_a} + \frac{1}{m_b} + \frac{(x_a \times n)^2}{I_a} + \frac{(x_b \times n)^2}{I_b}}$$

всегда число > 0 !

Collision Detection

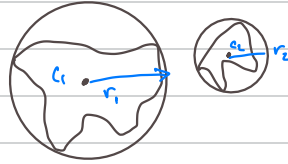
One sphere vs one sphere, even:

$$(c_1 - c_2)(c_1 - c_2) > (r_1 + r_2)^2$$

$$(c_1 - c_2)^2 > (r_1 + r_2)^2$$

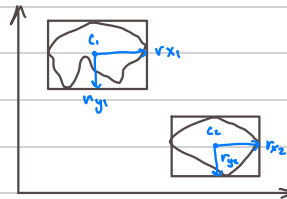
расстояние м/г
центров

сумма радиусов



r_x, r_y - radii

Axis Aligned Bounding Boxes AABB



AABB = 4-DOP

He nepecekaioce, even xoiu du oquo
sepuo:

$$|cx_1 - cx_2| > rx_1 + rx_2$$

$$|cy_1 - cy_2| > ry_1 + ry_2$$

Fluid Simulation

u - скорость

p - давление

ρ - плотность - mass per volume

Fluids flow from high to low pressure areas (along the gradient)

$$F_{\text{pressure}} = m \cdot \frac{1}{\rho} \nabla p = \tilde{V} \nabla p$$

↑
гравитация $\frac{m}{\rho}$, не меняется, а константа отмен

Направление к low pressure area

Viscosity - вязкость

опущенное
поле
↓ $\approx \tau, \chi$

одр., кот. показыват, насколько сильно
имеет место. завис. $\approx$
завис. от расстоян.
радиус отталкивания

$$A(x) \approx \sum_j A_j \frac{m_j}{\rho_j} W(x - x_j, h)$$

значение
поля
соседней
частицы

"объем"

расстояние до другой точки

объем сферы радиуса h

$$O(n N_h)$$

$$N_h = \frac{\text{Influence Domain}}{\text{Particle Domain}} = \frac{\frac{4}{3} \pi h^3}{\frac{4}{3} \pi r^3} = \frac{h^3}{r^3}$$

омиграное кон-но
соседних частиц

$$N_h = \frac{h^3}{r^3}$$

объем сферы
одной частицы
learn



Kernel should:

- Be positive everywhere
- Be symmetric
- Be spatially compact
- Be close enough to a Gaussian
- Integrate to 1

Pressure Gradient

∇p

1. Считаем мощность частиц:

$$p_i = \sum_j m_j w_{ij}$$

learn

2. Конвертируем мощность в давление:

$$p_i = \kappa \left(\left(\frac{p_i}{p_0} \right)^{\gamma} - 1 \right)$$

learn

Discretization

First order accurate:

• Forward differences:

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} + O(h)$$

→ truncation error

• Backward differences:

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1}))}{h} + O(h)$$

Second order accurate:

learn

• Central differences:

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h} + O(h^2)$$

• Second order derivative:

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2} + O(h^2)$$

$$(\text{grad } u)_{:j} = (\nabla u)_{:j} \approx \left(\frac{u_{i+1,j,k} - u_{i-1,j,k}}{2\Delta x}, \frac{u_{i,j,k} - u_{i,j-1,k}}{2\Delta y}, \frac{u_{i,j,k+1} - u_{i,j,k-1}}{2\Delta z} \right)$$

Divergence of vector field:

Gives scalar

$\nabla \cdot \mathbf{V} = \left(\frac{\partial}{\partial x} \quad \frac{\partial}{\partial y} \right) \cdot \mathbf{V}$ if $\mathbf{V} = \begin{pmatrix} u \\ v \end{pmatrix}$, then $\nabla \cdot \mathbf{V} = \left(\frac{\partial}{\partial x} \quad \frac{\partial}{\partial y} \right) \cdot \begin{pmatrix} u \\ v \end{pmatrix} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$

↗ actually means no x

Laplacian of a scalar field:

Second order operator

$$\Delta \phi = \nabla \cdot \nabla \phi = \left(\frac{\partial}{\partial x} \quad \frac{\partial}{\partial y} \quad \frac{\partial}{\partial z} \right) \cdot \left(\frac{\partial \phi}{\partial x} \quad \frac{\partial \phi}{\partial y} \quad \frac{\partial \phi}{\partial z} \right) = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$$

divergence gradient

Gradient of a function:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

Rotation matrices:

$$R_x = \begin{pmatrix} 1 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_y = \begin{pmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{pmatrix}$$

$$R_z = \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Translation Matrix:

$$T = \begin{pmatrix} 1 & 0 & 0 & \Delta x \\ 0 & 1 & 0 & \Delta y \\ 0 & 0 & 1 & \Delta z \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Euler:

$$\mathbf{x}^{[1]} = \mathbf{x}^{[0]} + \Delta t \cdot \mathbf{v}^{[0]}$$

$$\mathbf{v}^{[1]} = \mathbf{v}^{[0]} + \Delta t \cdot \mathbf{a}^{[0]}$$

Exercise 1

In a three-dimensional space, a mass point is at position $\mathbf{x} = (1, 0, 0)^T \text{ m}$ and moves at a constant velocity of $\mathbf{v} = (1, 1, 0)^T \frac{\text{m}}{\text{s}}$. Calculate its position after $\Delta t = 1 \text{ s}$.

$$\mathbf{x}^{[1]} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}_{\text{m}} + 1 \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}_{\frac{\text{m}}{\text{s}}} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}_{\text{m}}$$

Exercise 2

Another mass point moves with velocity $\mathbf{v} = (1, 1, 0)^T$. It experiences a constant acceleration of $\mathbf{a} = (2, -1, 0)^T \frac{\text{m}}{\text{s}^2}$. Calculate its velocity after $\Delta t = 0.1 \text{ s}$.

$$\mathbf{v}^{[1]} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}_{\frac{\text{m}}{\text{s}}} + 0.1 \text{ s} \cdot \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}_{\frac{\text{m}}{\text{s}^2}} = \begin{pmatrix} 1.2 \\ 0.9 \\ 0 \end{pmatrix}_{\frac{\text{m}}{\text{s}}}$$

Exercise 3

A mass point with mass $m_1 = 2 \text{ kg}$ and another mass point with mass $m_2 = 5 \text{ kg}$ both experience an identical force of $\mathbf{F} = (1, 2, 0)^T \text{ N}$. Calculate the corresponding acceleration of each mass point.

$$\mathbf{a}_1 = \frac{\mathbf{F}}{m_1} = \frac{(1, 2, 0)^T \text{ N}}{2 \text{ kg}} = \begin{pmatrix} 0.5 \\ 1 \\ 0 \end{pmatrix}_{\frac{\text{m}}{\text{s}^2}}$$

$$\mathbf{a}_2 = \frac{\mathbf{F}}{m_2} = \frac{(1, 2, 0)^T \text{ N}}{5 \text{ kg}} = \begin{pmatrix} 0.2 \\ 0.4 \\ 0 \end{pmatrix}_{\frac{\text{m}}{\text{s}^2}}$$

Exercise 4

A spring with a rest length of $L = 2 \text{ m}$ connects two mass points which are $l = 1 \text{ m}$ apart. The spring constant is $k = 10 \frac{\text{N}}{\text{m}}$. Which is the magnitude of the force experienced by each mass point? ($\|\mathbf{F}_1\|$ or $\|\mathbf{F}_2\|$)

$$\text{Magnitude - givena neposha} - \sqrt{x^2 + y^2} \quad \|\mathbf{F}_i\| = \sqrt{x_i^2 + y_i^2}$$

$$\|\mathbf{F}_2\| = \left\| -k(l-L) \frac{\mathbf{x}_2 - \mathbf{x}_1}{l} \right\| = |-k(l-L)| \cdot \left\| \frac{\mathbf{x}_2 - \mathbf{x}_1}{l} \right\| = |-k(l-L)| \cdot \frac{\|\mathbf{x}_2 - \mathbf{x}_1\|}{\|\mathbf{x}_2 - \mathbf{x}_1\|} = -k(l-L) = -10(1-2) = 10$$

модуль,
норму вектора
модуль
длина
отрезка.

→ это не что
модуль

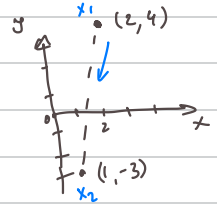
Exercise 5

[From now on, we will ignore units] Two mass points are connected with a spring. Compute their positions after two time steps using an Euler method, i.e., $\mathbf{x}_1^{[2]}$ and $\mathbf{x}_2^{[2]}$. The initial positions are $\mathbf{x}_1^{[0]} = (2, 4)^T$ and $\mathbf{x}_2^{[0]} = (1, -3)^T$. Initially, the velocity of both mass points is zero, $\mathbf{v}_1^{[0]} = \mathbf{v}_2^{[0]} = (0, 0)^T$, i.e., they are at rest. Both mass points have mass $m = 1$. The spring stiffness is $k = 2$. The rest length of the spring is $L = 4$. The time step is $\Delta t = 0.5$.

$$\mathbf{v}^{[0]} = 0, \mathbf{a}^{[0]} = 0, \mathbf{x}_1^{[0]} = (2, 4)^T, \mathbf{x}_2^{[0]} = (1, -3)^T, m_1 = 1, m_2 = 1, k = 2, L = 4, \Delta t = 0.5$$

$$1. F = m a$$

$$a_1^{[0]} = \frac{F_{1x}}{m_1} = \frac{-k(l-L) \frac{(x_1 - x_2)}{l}}{m_1} = \frac{-2(5\sqrt{2}-4) \cdot (\frac{1}{7})}{5\sqrt{2}} =$$



$$= \frac{(-10\sqrt{2}+8) \cdot (\frac{1}{7})}{5\sqrt{2}} = \frac{(14.142+8) \cdot (\frac{1}{7})}{7.071} = \frac{-6.142}{7.071} \cdot (\frac{1}{7}) =$$

$$= -0.869 \cdot (\frac{1}{7}) = \begin{pmatrix} -0.869 \\ -6.08 \end{pmatrix}$$

$$l = \|\mathbf{x}_2 - \mathbf{x}_1\| = \left\| \begin{pmatrix} 1 \\ -3 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \end{pmatrix} \right\| =$$

$$= \left\| \begin{pmatrix} -1 \\ -7 \end{pmatrix} \right\| = \sqrt{1+49} = \sqrt{50} = 5\sqrt{2}$$

$$\mathbf{F}_{12} = -\mathbf{F}_{21} = \begin{pmatrix} 0.869 \\ 6.08 \end{pmatrix}$$

$$\frac{F_{2x}}{F_1} = -k(l-L) \frac{x_1 - x_2}{l} = k(l-L) \frac{dx_2}{dt}$$

$$2. \mathbf{x}_1^{[1]} = \mathbf{x}_1^{[0]} + \Delta t \mathbf{v}_1^{[0]} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} + 0.5 \cdot 0 = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$\mathbf{x}_2^{[1]} = \begin{pmatrix} 1 \\ -3 \end{pmatrix} + 0.5 \cdot 0 = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

$$\mathbf{v}_1^{[1]} = \mathbf{v}_1^{[0]} + \Delta t \cdot \mathbf{a}_1^{[0]} = 0 + 0.5 \cdot \begin{pmatrix} 0.869 \\ -6.08 \end{pmatrix} = \begin{pmatrix} 0.4345 \\ -3.04 \end{pmatrix}$$

$$\mathbf{v}_2^{[1]} = \mathbf{v}_2^{[0]} + 0.5 \cdot \mathbf{a}_2^{[0]} = \begin{pmatrix} 0.4345 \\ 3.04 \end{pmatrix}$$



$$3. \mathbf{x}_1^{[2]} = \mathbf{x}_1^{[1]} + \Delta t \mathbf{v}_1^{[1]} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} + 0.5 \cdot \begin{pmatrix} 0.4345 \\ -3.04 \end{pmatrix} = \begin{pmatrix} 1.783 \\ 2.48 \end{pmatrix}$$

$$\mathbf{x}_2^{[2]} = \mathbf{x}_2^{[1]} + \Delta t \mathbf{v}_2^{[1]} = \begin{pmatrix} 1 \\ -3 \end{pmatrix} + 0.5 \cdot \begin{pmatrix} 0.4345 \\ 3.04 \end{pmatrix} = \begin{pmatrix} 1.217 \\ -2.518 \end{pmatrix}$$

$$x_2 - x_1$$

$$\mathbf{F}_{21} = -k(l-L) \frac{x_2 - x_1}{l}$$

Exercise 6

Again, two mass points are connected with a spring. Their initial positions are $\mathbf{x}_1^{[0]} = (-2, 2)^T$ and $\mathbf{x}_2^{[0]} = (3, 2)^T$ with initial velocities $\mathbf{v}_1^{[0]} = (-1, -1.5)^T$ and $\mathbf{v}_2^{[0]} = (1, -1.5)^T$. Their masses are different with $m_1 = 2$ and $m_2 = 3$. The spring stiffness is $k = 4$ and the rest length is $L = 6$. The time step is $\Delta t = 0.5$. Additionally, the mass points are subject to an acceleration due to gravity of $\mathbf{g} = (0, -10)^T$. Use an Euler method to compute the positions after two time steps, i.e., $\mathbf{x}_1^{[2]}$ and $\mathbf{x}_2^{[2]}$.

$$1. \mathbf{F}_1^{[0]} = m_1 \mathbf{a}_1^{[0]}$$

$$\mathbf{F}_1^{[0]} = \mathbf{F}_{12}^{[0]} + \mathbf{F}_{gr} \quad \mathbf{F}_{gr} = m_1 \mathbf{g}$$

$$\mathbf{F}_{12}^{[0]} = -k(l-L) \frac{\mathbf{x}_1^{[0]} - \mathbf{x}_2^{[0]}}{l} \quad l = \|\mathbf{x}_1^{[0]} - \mathbf{x}_2^{[0]}\| = \left\| \begin{pmatrix} -2 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \end{pmatrix} \right\| = \left\| \begin{pmatrix} -5 \\ 0 \end{pmatrix} \right\| = \sqrt{5^2} = 5$$

$$\mathbf{F}_{12}^{[0]} = -4(5-6) \frac{\begin{pmatrix} -2 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \end{pmatrix}}{5} = \frac{4}{5} \begin{pmatrix} -5 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 0 \end{pmatrix}$$

$$\mathbf{F}_1^{[0]} = \begin{pmatrix} -4 \\ 0 \end{pmatrix} + 2 \cdot \begin{pmatrix} 0 \\ -10 \end{pmatrix} = \begin{pmatrix} -4 \\ -20 \end{pmatrix}$$

$$\mathbf{a}_1^{[0]} = \frac{\mathbf{F}_1^{[0]}}{m_1} = \begin{pmatrix} -4 \\ -20 \end{pmatrix} : 2 = \begin{pmatrix} -2 \\ -10 \end{pmatrix}$$

$$2. \mathbf{F}_2^{[0]} = m_2 \mathbf{a}_2^{[0]}$$

$$\mathbf{F}_2^{[0]} = \mathbf{F}_{21}^{[0]} + \mathbf{F}_{gr} = -\mathbf{F}_{12}^{[0]} + m_2 \mathbf{g} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} + 3 \cdot \begin{pmatrix} 0 \\ -10 \end{pmatrix} = \begin{pmatrix} 4 \\ -30 \end{pmatrix}$$

$$\mathbf{a}_2^{[0]} = \frac{\mathbf{F}_2^{[0]}}{m_2} = \begin{pmatrix} 4 \\ -30 \end{pmatrix} : 3 = \begin{pmatrix} \frac{4}{3} \\ -10 \end{pmatrix}$$

$$3. \mathbf{x}_1^{[1]} = \mathbf{x}_1^{[0]} + \Delta t \mathbf{v}_1^{[0]} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} + 0.5 \cdot \begin{pmatrix} -1 \\ -1.5 \end{pmatrix} = \begin{pmatrix} -2.5 \\ 1.25 \end{pmatrix}$$

$$\mathbf{x}_2^{[1]} = \mathbf{x}_2^{[0]} + \Delta t \mathbf{v}_2^{[0]} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} + 0.5 \cdot \begin{pmatrix} 1 \\ -1.5 \end{pmatrix} = \begin{pmatrix} 3.5 \\ 1.25 \end{pmatrix}$$

$$\mathbf{v}_1^{[1]} = \mathbf{v}_1^{[0]} + \Delta t \mathbf{a}_1^{[0]} = \begin{pmatrix} -1 \\ -1.5 \end{pmatrix} + 0.5 \cdot \begin{pmatrix} -2 \\ -10 \end{pmatrix} = \begin{pmatrix} -2 \\ -6.5 \end{pmatrix}$$

$$\mathbf{v}_2^{[1]} = \mathbf{v}_2^{[0]} + \Delta t \mathbf{a}_2^{[0]} = \begin{pmatrix} 1 \\ -1.5 \end{pmatrix} + 0.5 \cdot \begin{pmatrix} \frac{4}{3} \\ -10 \end{pmatrix} = \begin{pmatrix} 1 \\ -1.5 \end{pmatrix} + \begin{pmatrix} \frac{2}{3} \\ -5 \end{pmatrix} = \begin{pmatrix} \frac{5}{3} \\ -6.5 \end{pmatrix}$$

$$4. \mathbf{x}_1^{[2]} = \mathbf{x}_1^{[1]} + \Delta t \mathbf{v}_1^{[1]} = \begin{pmatrix} -2.5 \\ 1.25 \end{pmatrix} + 0.5 \cdot \begin{pmatrix} -2 \\ -6.5 \end{pmatrix} = \begin{pmatrix} -3.5 \\ -2 \end{pmatrix}$$

$$\mathbf{x}_2^{[2]} = \mathbf{x}_2^{[1]} + \Delta t \mathbf{v}_2^{[1]} = \begin{pmatrix} 3.5 \\ 1.25 \end{pmatrix} + 0.5 \cdot \begin{pmatrix} \frac{5}{3} \\ -6.5 \end{pmatrix} = \begin{pmatrix} 3.5 \\ 1.25 \end{pmatrix} + \begin{pmatrix} \frac{2.5}{3} \\ -3.25 \end{pmatrix} = \begin{pmatrix} 4.33 \\ -2 \end{pmatrix}$$

Exercise 7

(From Retake Exam 2023/2024)

A mass spring system in 2d consists of three mass points and two springs in the following connectivity: point 1 is connected to point 2, point 2 is connected to point 3. There is **no** connection between point 1 and point 3. All springs have the same stiffness $k = 2$ and the same (unknown) rest length L . Point 1 is at location $\mathbf{x}_1 = (2, 3)$. The unit direction vector pointing from point 1 to point 2 is $\mathbf{d}_{1 \rightarrow 2} = (1, 0)$, ~~and the unit direction vector pointing from point 2 to point 3 is $\mathbf{d}_{2 \rightarrow 3} = (1, 0)$~~ . Point 1 is $l_{1 \rightarrow 2} = 2$ apart from point 2. Point 2 is $l_{2 \rightarrow 3} = 5$ apart from point 3. The total force experienced by point 1 is $\mathbf{F}_1 = (3, 0)$, and by point 2 is $\mathbf{F}_2 = (2, 2)$. The total force acting on point 3 is unknown. Compute the following:

1. The position of point 2.
2. Use the knowledge of the total force acting on point 1 to determine the spring's rest length L .
3. The total force acting on point 3.



$$1) \mathbf{x}_2 = \mathbf{x}_1 + \mathbf{d}_{1 \rightarrow 2} \cdot l = \begin{pmatrix} 2 \\ 3 \end{pmatrix} + 2 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$2) \mathbf{F}_1 = -k(l_n - L) \frac{\mathbf{x}_1 - \mathbf{x}_2}{l_{12}}$$

$$\frac{\mathbf{F}_1 l_{12}}{-k(\mathbf{x}_1 - \mathbf{x}_2)} = l_n - L$$

$$L_n = l_n - \frac{\mathbf{F}_1 l_{12}}{-k(\mathbf{x}_1 - \mathbf{x}_2)} = 2 + \frac{\begin{pmatrix} 3 \\ 0 \end{pmatrix} \cdot 2}{2 \left(\begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 4 \\ 3 \end{pmatrix} \right)} = 2 + \frac{\begin{pmatrix} 3 \\ 0 \end{pmatrix}}{\begin{pmatrix} -2 \\ 0 \end{pmatrix}} = 2 + (-1.5) = 0.5$$

$$3) \mathbf{F}_3 = -\mathbf{F}_{23}$$

$$\mathbf{F}_2 = \mathbf{F}_{21} + \mathbf{F}_{23} = -\mathbf{F}_1 + \mathbf{F}_{23}$$

$$\mathbf{F}_{23} = \mathbf{F}_2 + \mathbf{F}_1 = \begin{pmatrix} 2 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$$

$$\mathbf{F}_3 = -\begin{pmatrix} 5 \\ 2 \end{pmatrix} = \begin{pmatrix} -5 \\ -2 \end{pmatrix}$$

Exercise 1

Integrate the following ordinary differential equation (ODE):

$$\frac{dy}{dt} = -y$$

for 1 step using $\Delta t = 0.1$, starting from $y(0) = 1$, using

1. an explicit Euler scheme
2. Heun's method
3. the midpoint method
4. (optional) the Runge-Kutta-4 method

← *npaywagayro nagpamali sa*

$$1 \quad f(t, y) = -y. \text{ It doesn't depend on } t, \text{ so } f(y) = -y$$

$$y^{n+1} = y^n + \Delta t \cdot f(y^n) \leftarrow \text{npaywagayro } f(y) = -y$$

$$y^{n+1} = 1 + 0.1 \cdot (-1) = 0.9$$

2. Use Euler step $\tilde{y} = 0.9$

$$y^{[1]} = y^{[0]} + \frac{\Delta t}{2} \cdot (f(y^{[0]}) + f(\tilde{y})) = 1 + 0.05(-1 + (-0.9)) = 1 + 0.05(-1.9) = 0.905$$

$$3. \tilde{y} = y^{[0]} + \frac{\Delta t}{2} f(y^{[0]}) = 1 + 0.05(-1) = 1 - 0.05 = 0.95$$

$$y^{[1]} = y^{[0]} + \Delta t f(\tilde{y}) = 1 + 0.1 \cdot (-0.95) = 0.905$$

т.е. we jump on Δt

$$4. k_1 = f(y^{[0]}) = -1$$

$$k_2 = f(y^{[0]} + \frac{\Delta t}{2} k_1) = f(1 + 0.05 \cdot (-1)) = f(0.95) = -0.95$$

$$k_3 = f(y^{[0]} + \frac{\Delta t}{2} k_2) = f(1 + 0.05 \cdot (-0.95)) = f(0.9525) = -0.9525$$

$$k_4 = f(y^{[0]} + \Delta t \cdot k_3) = f(1 + 0.1 \cdot (-0.9525)) = -0.90475$$

$$y^{[1]} = y^{[0]} + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4) = 1 + \frac{0.1}{6} (-1 + 2 \cdot (-0.95) + 2 \cdot (-0.9525) - 0.90475) =$$

$$= 1 + \frac{0.1}{6} (-1 - 1.9 - 1.905 - 0.90475) = 1 - \frac{0.1 \cdot 5.70975}{6} = 0.90475$$

Exercise 2

Two mass points are connected with a spring. Their initial positions are $\mathbf{x}_1^{[0]} = (-2, 2)^T$ and $\mathbf{x}_2^{[0]} = (3, 2)^T$ with initial velocities $\mathbf{v}_1^{[0]} = (-1, -1.5)^T$ and $\mathbf{v}_2^{[0]} = (1, -1.5)^T$. Their masses are different with $m_1 = 2$ and $m_2 = 3$. The spring stiffness is $k = 4$ and the rest length is $L = 6$. The time step is $\Delta t = 0.5$. Additionally, the mass points are subject to an acceleration due to gravity of $\mathbf{g} = (0, -10)^T$. Compute the positions and velocities after one time step, i.e., $\mathbf{x}_1^{[1]}$, $\mathbf{x}_2^{[1]}$, $\mathbf{v}_1^{[1]}$, and $\mathbf{v}_2^{[1]}$, using

1. Heun's method

2. the midpoint method

3. Leap-Frog (assume that the initial velocity is given as $\mathbf{v}_1^{[-1/2]}$ and $\mathbf{v}_2^{[-1/2]}$, and the desired velocity to be computed shall be $\mathbf{v}_1^{[1/2]}$ and $\mathbf{v}_2^{[1/2]}$)

$$1. \mathbf{F}_i = m_i \mathbf{a}_i^{[0]}$$

$$\mathbf{F}_i = \mathbf{F}_{12} + \mathbf{F}_{gr} \quad \mathbf{F}_{gr} = m_i \mathbf{g}$$

$$\mathbf{F}_{12} = -k(l-L) \frac{\mathbf{x}_2^{[0]} - \mathbf{x}_1^{[0]}}{l} \quad l = \|\mathbf{x}_1^{[0]} - \mathbf{x}_2^{[0]}\| = \left\| \begin{pmatrix} -2 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \end{pmatrix} \right\| = \left\| \begin{pmatrix} -5 \\ 0 \end{pmatrix} \right\| = \sqrt{5^2} = 5$$

$$\mathbf{F}_{12} = -4(5-6) \frac{\begin{pmatrix} 3 \\ 2 \end{pmatrix} - \begin{pmatrix} -2 \\ 2 \end{pmatrix}}{5} = \frac{4}{5} \begin{pmatrix} -5 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 0 \end{pmatrix}$$

$$\mathbf{F}_1 = \begin{pmatrix} -4 \\ 0 \end{pmatrix} + 2 \cdot \begin{pmatrix} 0 \\ -10 \end{pmatrix} = \begin{pmatrix} -4 \\ -20 \end{pmatrix}$$

$$\mathbf{a}_1^{[0]} = \frac{\mathbf{F}_1^{[0]}}{m_1} = \begin{pmatrix} -4 \\ -20 \end{pmatrix} : 2 = \begin{pmatrix} -2 \\ -10 \end{pmatrix}$$

$$2. \mathbf{F}_2^{[0]} = m_2 \mathbf{a}_2^{[0]}$$

$$\mathbf{F}_2 = \mathbf{F}_{21} + \mathbf{F}_{gr} = -\mathbf{F}_{12} + m_2 \cdot \mathbf{g} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} + 3 \cdot \begin{pmatrix} 0 \\ -10 \end{pmatrix} = \begin{pmatrix} 4 \\ -30 \end{pmatrix}$$

$$\mathbf{a}_2^{[0]} = \frac{\mathbf{F}_2^{[0]}}{m_2} = \begin{pmatrix} 4 \\ -30 \end{pmatrix} : 3 = \begin{pmatrix} \frac{4}{3} \\ -10 \end{pmatrix}$$

1) 1. Euler step:

$$\tilde{x}_1 = x_1^{[0]} + \Delta t \cdot v_1^{[0]} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} + 0.5 \cdot \begin{pmatrix} -1 \\ -1.5 \end{pmatrix} = \begin{pmatrix} -2.5 \\ 1.25 \end{pmatrix}$$

$$\tilde{x}_2 = x_2^{[0]} + \Delta t \cdot v_2^{[0]} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} + 0.5 \cdot \begin{pmatrix} -1 \\ -1.5 \end{pmatrix} = \begin{pmatrix} 1.5 \\ 1.25 \end{pmatrix}$$

$$\tilde{v}_1 = v_1^{[0]} + \Delta t \cdot a_1^{[0]} = \begin{pmatrix} -1 \\ -1.5 \end{pmatrix} + 0.5 \cdot \begin{pmatrix} -2 \\ -10 \end{pmatrix} = \begin{pmatrix} -2 \\ -6.5 \end{pmatrix}$$

$$\tilde{v}_2 = v_2^{[0]} + \Delta t \cdot a_2^{[0]} = \begin{pmatrix} -1.5 \\ -10 \end{pmatrix} + 0.5 \cdot \begin{pmatrix} 4 \\ -5 \end{pmatrix} = \begin{pmatrix} -1.5 \\ -10 \end{pmatrix} + \begin{pmatrix} 2 \\ -2.5 \end{pmatrix} = \begin{pmatrix} 0.5 \\ -12.5 \end{pmatrix}$$

2. Runge method:

$$1. x_1^{[1]} = x_1^{[0]} + \frac{\Delta t}{2} (v_1^{[0]} + \tilde{v}_1) = \begin{pmatrix} -2 \\ 2 \end{pmatrix} + \frac{0.5}{2} \left(\begin{pmatrix} -1 \\ -1.5 \end{pmatrix} + \begin{pmatrix} -2 \\ -6.5 \end{pmatrix} \right) = \begin{pmatrix} -2 \\ 2 \end{pmatrix} + \frac{0.5}{2} \begin{pmatrix} -3 \\ -8 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} + \begin{pmatrix} -\frac{3}{4} \\ -1 \end{pmatrix} = \begin{pmatrix} -\frac{11}{4} \\ 1 \end{pmatrix}$$

$$x_2^{[1]} = x_2^{[0]} + \frac{\Delta t}{2} (v_2^{[0]} + \tilde{v}_2) = \begin{pmatrix} 2 \\ 2 \end{pmatrix} + \frac{0.5}{2} \left(\begin{pmatrix} -1.5 \\ -10 \end{pmatrix} + \begin{pmatrix} 0.5 \\ -12.5 \end{pmatrix} \right) = \begin{pmatrix} 2 \\ 2 \end{pmatrix} + \frac{1}{4} \cdot \begin{pmatrix} -8 \\ -22.5 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} + \begin{pmatrix} -2 \\ -5.625 \end{pmatrix} = \begin{pmatrix} 0 \\ -3.625 \end{pmatrix}$$

2. Если $a_1^{[0]}$ и $a_2^{[0]}$ известны, то $\tilde{a}_1$ и $\tilde{a}_2$.

$\tilde{a}_1$ и $\tilde{a}_2$ - это Euler step, т.е. просто ускорение в следующий момент времени по Euler.

Возьмем точки и пересчитаем для них F. Для этого нужно найти ℓ .

$$\ell = \left\| \begin{pmatrix} -2.5 \\ 1.25 \end{pmatrix} - \begin{pmatrix} 1.5 \\ 1.25 \end{pmatrix} \right\| = \left\| \begin{pmatrix} -4 \\ 0 \end{pmatrix} \right\| = \sqrt{16} = 4.$$

$$L = 6 \text{ и } \ell = 4$$

$$F = 0, \text{ т.к. } \ell - L = 0$$

Тогда $\tilde{a}_1 = \tilde{a}_2 = \begin{pmatrix} 0 \\ -10 \end{pmatrix}$ - у нас есть уравнение

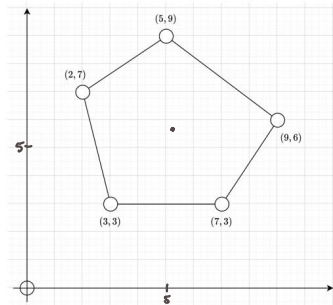
$$v_1^{[1]} = v_1^{[0]} + \frac{\Delta t}{2} (a_1^{[0]} + \tilde{a}_1) = \begin{pmatrix} -1 \\ -1.5 \end{pmatrix} + \frac{1}{4} \begin{pmatrix} -2 \\ -10 \end{pmatrix} = \begin{pmatrix} -1 \\ -1.5 \end{pmatrix} - \begin{pmatrix} 0.5 \\ 2.5 \end{pmatrix} = \begin{pmatrix} -1.5 \\ -4 \end{pmatrix}$$

$$v_2^{[1]} = v_2^{[0]} + \frac{\Delta t}{2} (a_2^{[0]} + \tilde{a}_2) = \begin{pmatrix} -1.5 \\ -10 \end{pmatrix} + \frac{1}{4} \begin{pmatrix} 4 \\ -5 \end{pmatrix} = \begin{pmatrix} -1.5 \\ -10 \end{pmatrix} + \begin{pmatrix} 1 \\ -1.25 \end{pmatrix} = \begin{pmatrix} -0.5 \\ -11.25 \end{pmatrix}$$

Задана функция

Exercise 1

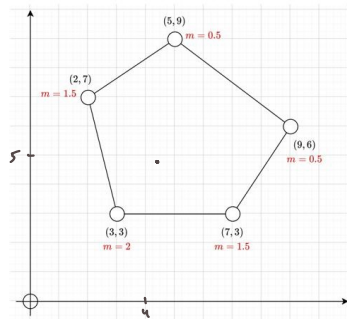
Compute the center of mass for the rigid body shown below. Each vertex of the rigid body has the same mass m .



$$\frac{1.5 \begin{pmatrix} 2 \\ 7 \end{pmatrix} + 0.5 \begin{pmatrix} 5 \\ 9 \end{pmatrix} + 0.5 \begin{pmatrix} 9 \\ 6 \end{pmatrix} + 1.5 \begin{pmatrix} 7 \\ 3 \end{pmatrix} + 2 \begin{pmatrix} 3 \\ 3 \end{pmatrix}}{1.5 + 0.5 + 0.5 + 1.5 + 2} = \frac{\begin{pmatrix} 26 \\ 20 \end{pmatrix}}{6} = \begin{pmatrix} 4.33 \\ 3.33 \end{pmatrix}$$

Exercise 2

The same rigid body as in the previous exercise has now been modified such that each point has an individual weight as shown below. Compute the center of mass.



$$\frac{1.5 \begin{pmatrix} 2 \\ 7 \end{pmatrix} + 0.5 \begin{pmatrix} 5 \\ 9 \end{pmatrix} + 0.5 \begin{pmatrix} 9 \\ 6 \end{pmatrix} + 1.5 \begin{pmatrix} 7 \\ 3 \end{pmatrix} + 2 \begin{pmatrix} 3 \\ 3 \end{pmatrix}}{1.5 + 0.5 + 0.5 + 1.5 + 2} = \frac{\begin{pmatrix} 3 \\ 10.5 \end{pmatrix} + \begin{pmatrix} 2.5 \\ 4.5 \end{pmatrix} + \begin{pmatrix} 4.5 \\ 3 \end{pmatrix} + \begin{pmatrix} 10.5 \\ 4.5 \end{pmatrix} + \begin{pmatrix} 6 \\ 6 \end{pmatrix}}{6} = \begin{pmatrix} 4.33 \\ 3.33 \end{pmatrix}$$

Exercise 3

Compute the mass moment of inertia for the first rigid body. (Optional: Also compute the mass moment of inertia for the second rigid body with different weights for each point.)

$$I = \sum_n m_n x_n \cdot x_n \quad m_n = 1$$

But we need to calculate relative coordinates first.

$$cm = \begin{pmatrix} 5.2 \\ 5.6 \end{pmatrix}$$

$$I = (x_1 \cdot x_1 + x_2 \cdot x_2 + x_3 \cdot x_3 + x_4 \cdot x_4 + x_5 \cdot x_5)$$

$$x_1 = \begin{pmatrix} 2 \\ 7 \end{pmatrix} - \begin{pmatrix} 5.2 \\ 5.6 \end{pmatrix} = \begin{pmatrix} -3.2 \\ 2.6 \end{pmatrix}$$

$$x_1 \cdot x_1 = (-3.2)^2 + (2.6)^2$$



$$|a||b|\cos\alpha$$

$$x_2 = \begin{pmatrix} 5 \\ 9 \end{pmatrix} - \begin{pmatrix} 5.2 \\ 5.6 \end{pmatrix} = \begin{pmatrix} -0.2 \\ 3.4 \end{pmatrix}$$

$$x_3 = \begin{pmatrix} 9 \\ 6 \end{pmatrix} - \begin{pmatrix} 5.2 \\ 5.6 \end{pmatrix} = \begin{pmatrix} 3.8 \\ 0.4 \end{pmatrix}$$

$$x_4 = \begin{pmatrix} 7 \\ 3 \end{pmatrix} - \begin{pmatrix} 5.2 \\ 5.6 \end{pmatrix} = \begin{pmatrix} 1.8 \\ -2.6 \end{pmatrix}$$

$$x_5 = \begin{pmatrix} 3 \\ 3 \end{pmatrix} - \begin{pmatrix} 5.2 \\ 5.6 \end{pmatrix} = \begin{pmatrix} -2.2 \\ -2.6 \end{pmatrix}$$

Exercise 4

Compute the cross product between the vectors $\mathbf{a} = [1 \ 2 \ 3]^T$ and $\mathbf{b} = [4 \ 5 \ 6]^T$.

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 2 \cdot 6 - 3 \cdot 5 \\ 3 \cdot 4 - 1 \cdot 6 \\ 1 \cdot 5 - 2 \cdot 4 \end{pmatrix} = \begin{pmatrix} 12 - 15 \\ 12 - 6 \\ 5 - 8 \end{pmatrix} = \begin{pmatrix} -3 \\ 6 \\ -3 \end{pmatrix}$$

Exercise 5

Compute the cross product between the two 2D vectors $\mathbf{a} = [1 \ 2]^T$ and $\mathbf{b} = [3 \ 4]^T$. (Select the paragraphs with your mouse to reveal the hint.)

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \times \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \cdot 0 - 0 \cdot 4 \\ 0 \cdot 3 - 1 \cdot 0 \\ 1 \cdot 4 - 2 \cdot 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -2 \end{pmatrix} \quad \begin{matrix} a_1 b_2 - a_2 b_1 \\ \text{т.к. мы в 2D, то 3D считаем} \end{matrix}$$

Exercise 6

Compute the cross product between the scalar $s = -7$ and the 2D vector $\mathbf{a} = [1 \ 2]^T$. (Select the paragraphs with your mouse to reveal the hint.)

$$s \times \mathbf{a} = -7 \times \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -7 \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \cdot 0 - (-7) \cdot 2 \\ -7 \cdot 1 - 0 \cdot 0 \\ 0 \cdot 0 - 0 \cdot 0 \end{pmatrix} = \begin{pmatrix} 14 \\ -7 \\ 0 \end{pmatrix}$$

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Exercise 7

A rigid body in 2D rotates around its center of mass with an angular velocity $w^{[0]} = 4.0$ deg/s. The initial rotation is $r^{[0]} = 45$ deg. It has an inertia of $i = 5$ kgm²/deg. A force of $\mathbf{f} = [0 \ -20]^T$ $\frac{\text{kgm}}{\text{s}^2}$ is applied at a position of $\mathbf{x}_{\text{rel}} = [1 \ 0.5]^T$ m relative to the center of mass. Compute the updated angular velocity $w^{[1]}$ and the updated rotation $r^{[1]}$ after an explicit Euler time step of $\Delta t = 0.25$ s.

$$q = i \cdot a \quad \leftarrow \text{angular acceter}$$

$\nwarrow$ inertia

$$a = \frac{q}{i}$$

$$F = m a$$



$$q = \mathbf{x}_{\text{rel}} \times \mathbf{F} = \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} \times \begin{pmatrix} 0 \\ -20 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -20 \end{pmatrix} = -20$$

First update the position:

$$r^{[1]} = r^{[0]} + w^{[0]} \cdot \Delta t = 45 + 4 \cdot 0.25 = 46$$

$$w^{[1]} = w^{[0]} + \Delta t \cdot \frac{q}{i} = 4 + 0.25 \cdot \frac{(-20)}{5} = 3$$

$$w_i^{[1]} = w_i^{[0]} + \Delta t a_{\text{rot}}$$

$$q = i \cdot a_{\text{rot}}$$

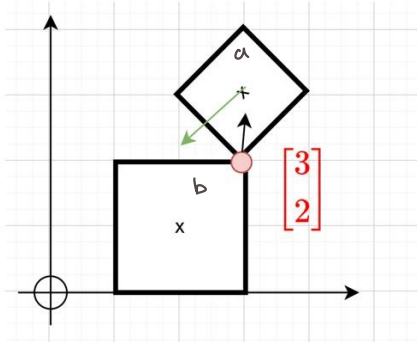
$$q = \mathbf{x}_{\text{rel}} \times \mathbf{F}$$

$$r = w \cdot \Delta t - \text{ypon, } \omega \text{on. } \eta \text{poyau}$$

$$v_{\text{na uppy}} = \frac{r \cdot w \cdot \Delta t}{\Delta t} = r \cdot w$$

Exercise 8

Compute the impulse J for the following collision and apply it to the two rigid bodies.



The center of mass of the two rigid bodies is at $\mathbf{x}_{cm,a} = [3 \ 3]^T$ and $\mathbf{x}_{cm,b} = [2 \ 1]^T$. Their masses and moments of inertia are $m_a = 4$, $m_b = 8$, $I_a = 6$, and $I_b = 8$. The linear and angular velocities are $\mathbf{v}_a = [-1 \ -1]^T$, $\mathbf{v}_b = [0 \ 0]^T$, $\omega_a = -4$, and $\omega_b = 0$. The collision happens at $\mathbf{x}_c = [3 \ 2]^T$ with a normal vector $\mathbf{n} = [0 \ 1]^T$. The coefficient of restitution is $c = 1.0$.

1. Найдем позицию вектор из $\mathbf{cm}_a$ в collision point и из $\mathbf{cm}_b$ в col. point:

$$\mathbf{x}_{ca} = \mathbf{x}_c - \mathbf{x}_{cm,a} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

$$\mathbf{x}_{cb} = \mathbf{x}_c - \mathbf{x}_{cm,b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

2. Найдем скорость точки в каждом теле в месте коллизии в момент коллизии:

$$\mathbf{v}_{ca} = \underbrace{\mathbf{v}_a}_{\text{скорость центра}} + \underbrace{\omega_a \times \mathbf{x}_{ca}}_{\text{угл. скор. объекта}} = \text{позиция вектор точки}$$

$$\mathbf{v} = \omega \times \mathbf{r}$$

но определяем радиуса
 $\downarrow$
 $s = \theta r$, где s — это

длина ка окруж.
 v — скорость края,
 θ — угол в радианах

$$= \begin{pmatrix} -1 \\ -1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -4 \end{pmatrix} \times \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 - (-4)(-1) \\ 0 - 0 \\ 0 - 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix} + \begin{pmatrix} -4 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -5 \\ -1 \\ 0 \end{pmatrix}$$

$$\mathbf{v}_{cb} = \mathbf{v}_b + \omega_b \times \mathbf{x}_{cb} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

θ — угол в радианах

 Если радиус — угол θ , то есть, то $s = r \theta$
 $\theta = \frac{s}{r}$

$$3. \mathbf{v}_{rel} = \mathbf{v}_{ca} - \mathbf{v}_{cb} = \begin{pmatrix} -5 \\ -1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -5 \\ -1 \\ 0 \end{pmatrix}$$

$$4. J = \frac{-1 + c}{\frac{1}{m_a} + \frac{1}{m_b} + \frac{(\mathbf{x}_a \times \mathbf{n})^2}{I_a} + \frac{(\mathbf{x}_b \times \mathbf{n})^2}{I_b}} \cdot \mathbf{v}_{rel} \cdot \mathbf{n} = \frac{-1 + 1}{\frac{1}{4} + \frac{1}{8} + \frac{(\begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix})^2}{6} + \frac{(\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix})^2}{8}} \cdot \begin{pmatrix} -5 \\ -1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \frac{-2(0-1)}{\frac{2}{8} + 0 + \frac{1}{8}} = \frac{2}{\frac{4}{8}} = \frac{2}{0.5} = 4.$$

$\Rightarrow 2\pi = 360^\circ$
 в радианах $\Rightarrow s = 2\pi r$ — длина окружн.

$$5. \mathbf{v}'_a = \mathbf{v}_a + \frac{J}{m_a} \mathbf{n} = \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix} + \frac{4}{4} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

$$\mathbf{v}'_b = \mathbf{v}_b - \frac{J}{m_b} \mathbf{n} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} - \frac{4}{8} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0.5 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -0.5 \\ 0 \end{pmatrix}$$

Update ω also

Problem 2 Rigid-Body Forces and Velocities (14 credits)

Consider the rigid body R in two dimensions, with the following world space state: center of mass $\mathbf{x}_c = (6.0, 2.0)$, linear velocity $\mathbf{v}_c = (3.0, 1.0)$, angular velocity $w_c = 4.0$, mass $m = 2.0$, and an inertia tensor $I = 3.0$. The geometry of the body is irrelevant, and you can ignore orientation as well as gravity.

a) An unknown force $\mathbf{f}$ is acting upon the rigid body at position $\mathbf{x}_f = (6.0, 4.0)$, in world space. After a time step of length $h = 0.2$, using Euler integration, the resulting linear velocity becomes $\mathbf{v}'_c = (8.0, 4.0)$. Calculate the force $\mathbf{f}$.

$$\mathbf{v}_c^{[1]} = \mathbf{v}_c^{[0]} + h \cdot \mathbf{a}$$

$$\mathbf{a} = \frac{\mathbf{v}_c^{[1]} - \mathbf{v}_c^{[0]}}{h}$$

$$\mathbf{F} = m\mathbf{a}$$

Т.е. по линейное движение, нам нужно считать суммарно все силы, действующие на объект (а она одна $\mathbf{f}$).

b) Assume a force $\mathbf{f} = (15.0, 0.0)$ is acting at $\mathbf{x}_f = (6.0, 4.0)$. Calculate resulting angular velocity (w'_c) for a time step of length $h = 0.2$, using Euler integration.

$$w_c^{[1]} = w_c^{[0]} + h \cdot a_{\text{rot}}$$

$$a_{\text{rot}} = \frac{\tau}{I} \quad \tau = \mathbf{x}_{\text{rel},f} \times \mathbf{f}$$

$$w_c^{[1]} = w_c^{[0]} + \frac{h(\mathbf{x}_{\text{rel},f} \times \mathbf{f})}{I} = 4 + \frac{0.2((2) \times (15))}{3} = 4 + \frac{0.2(0-30)}{3} = 4 - 2 = 2.$$

$$\mathbf{x}_{\text{rel},f} = \mathbf{x}_{w,f} - \mathbf{x}_c = \begin{pmatrix} 6 \\ 4 \end{pmatrix} - \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

c) Assume the rigid body R now has an angular momentum $L = 3$, an unknown linear velocity denoted with $\mathbf{v}_u$, and no forces are acting on the body. The velocity $\mathbf{v}_1$ of the point $\mathbf{x}_1 = (3, 2)$, on R is $\mathbf{v}_1 = (-2, 2)$. Both are world space coordinates. The other parameters of the rigid body are: $\mathbf{x}_c = (6.0, 2.0)$, $m = 2.0$, $I = 3.0$. Calculate the linear velocity $\mathbf{v}_u$ of the rigid body.

$$L = I\omega$$

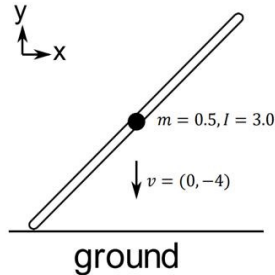
$$\mathbf{x}_{1,\text{rel}} = \mathbf{x}_1 - \mathbf{x}_c = \begin{pmatrix} 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \end{pmatrix}$$

$$\mathbf{v}_1^{\text{world}} = \mathbf{v}_{\text{cm}} + \omega \times \mathbf{x}_{1,\text{rel}}$$

$$\omega = \frac{L}{I} = \frac{3}{3} = 1$$

$$\mathbf{v}_u = \mathbf{v}_1^{\text{world}} - \omega \times \mathbf{x}_{1,\text{rel}} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} \times \begin{pmatrix} -3 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} - \begin{pmatrix} 0-0 \\ 1 \cdot (-3) - 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ -3 \end{pmatrix} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$$

Problem 6 Collisions and Impulse (11 credits)



A rigid rod in 2D (see figure above) falls onto the ground with a linear velocity $\mathbf{v} = (0.0, -4.0)$ and an angular velocity $\mathbf{w} = 2.0$. The collision happens at world space position $(-2.0, -2.0)$, and the center of mass of the rod is at world space position $(1.0, 1.0)$. The properties of the rod are: mass $m = 0.5$ and 2D inertia tensor $I = 3.0$. The coefficient of restitution is 0.5 .

- a) Calculate the resulting impulse J using the normal of the ground plane $\mathbf{n} = (0.0, 1.0)$.

$$\mathbf{x}_{rel,c} = \begin{pmatrix} -2 \\ -2 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \end{pmatrix}$$

$$\mathbf{v}_{rel,c} = \mathbf{v} + \mathbf{w} \times \mathbf{x}_{rel,c} = \begin{pmatrix} 0 \\ -4 \end{pmatrix} + \begin{pmatrix} 0 \\ 2 \end{pmatrix} \times \begin{pmatrix} -3 \\ -3 \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \end{pmatrix} + \begin{pmatrix} 0 - 2(-3) \\ 2(-3) - 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \end{pmatrix} + \begin{pmatrix} 6 \\ -6 \end{pmatrix} = \begin{pmatrix} 6 \\ -10 \end{pmatrix}$$

$$J = \frac{-(1+c) \mathbf{v}_{rel,c} \cdot \mathbf{n}}{\frac{1}{m} + \frac{(\mathbf{x}_{rel,c} \times \mathbf{n})^2}{I}} = \frac{-(1+0.5) \begin{pmatrix} 6 \\ -10 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix}}{\frac{1}{0.5} + \frac{\left(\begin{pmatrix} -3 \\ -3 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)^2}{3}} = \frac{-1.5 \cdot (-10)}{2 + \frac{\begin{pmatrix} -3 \cdot 0 - 0 \\ 0 - (-3) \cdot 0 \\ (-3) \cdot 1 - (-3) \cdot 0 \end{pmatrix}^2}{3}} = \frac{15}{2 + \frac{9}{3}} = \frac{15}{5} = 3.$$

- b) Now assume $J = 4.0$. Calculate the updated linear velocity $\mathbf{v}'$ and angular velocity $\mathbf{w}'$, respectively. Repeated from previous page: linear velocity $\mathbf{v} = (0.0, -4.0)$, angular velocity $\mathbf{w} = 2.0$, collision happens at $(-2.0, -2.0)$, and the center of mass $(1.0, 1.0)$, mass $m = 0.5$, 2D inertia tensor $I = 3.0$, coefficient of restitution 0.5 .

$$\mathbf{v}' = \mathbf{v} + \frac{J \cdot \mathbf{n}}{m} = \begin{pmatrix} 0 \\ -4 \end{pmatrix} + \frac{4 \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix}}{0.5} = \begin{pmatrix} 0 \\ -4 \end{pmatrix} + \begin{pmatrix} 0 \\ 8 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \end{pmatrix}$$

$$\mathbf{w}' = \mathbf{w} + \frac{\mathbf{x}_{rel,c} \times J \mathbf{n}}{I} = 2 + \frac{\begin{pmatrix} -3 \\ -3 \end{pmatrix} \times \begin{pmatrix} 0 \\ 4 \end{pmatrix}}{3} = 2 + \frac{\begin{pmatrix} 0 \\ -12 - 0 \end{pmatrix}}{3} = 2 + (-4) = -2.$$

Problem 8 SPH Interpolation (15 credits)

This task deals with Smoothed Particle Hydrodynamics (SPH) in one dimension with a kernel function

$$W(r, h) = \begin{cases} 1 - \left(\frac{|r|}{h}\right), & |r| \leq h \\ 0, & \text{else.} \end{cases} \quad (8.1)$$

a) Given a set of particle positions $x = [0, 1, 2, 2.5, 3]$ with equal support radius $h = \frac{7}{4}$ compute the adjacency matrix A such that each entry a_{ij} is 1 if particles i and j are neighbors and 0 if they are not. (Hint: it may be helpful to first make a list of signed or absolute distances between all particle pairs as a matrix. This is not required for points.)

$$h = \frac{7}{4} = 1.75$$

$$J = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 1 & 2 & 2.5 & 3 \\ 1 & 0 & 1 & 1.5 & 2 \\ 2 & 1 & 0 & 0.5 & 1 \\ 2.5 & 1.5 & 0.5 & 0 & 0.5 \\ 3 & 2 & 1 & 0.5 & 0 \end{bmatrix} \end{matrix} \quad A = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

b)* Given a particle system consisting of 6 particles of uniform mass $m = 2$ and support radius $h = 1$ with a particle distance matrix of

$$d_{ij} = \begin{bmatrix} 0 & 1 & 2 & 3.5 & 4 & 4.5 \\ 1 & 0 & 1 & 2.5 & 3 & 3.5 \\ 2 & 1 & 0 & 1.5 & 2 & 2.5 \\ 3.5 & 2.5 & 1.5 & 0 & 0.5 & 1 \\ 4 & 3 & 2 & 0.5 & 0 & 0.5 \\ 4.5 & 3.5 & 2.5 & 1 & 0.5 & 0 \end{bmatrix} \quad (8.4)$$

Compute the SPH-based summation particle density for all particles. Please provide answers as decimal values for ρ_1 through to ρ_6 .

$$W(r, h) = \begin{cases} 1 - |r|, & |r| \leq 1 \\ 0, & \text{else} \end{cases}$$

$$\rho_1 = 2(1-0) + 2 \cdot 0 = 2$$

$$\rho_2 = 2(1-1) + 2(1-0) + 2(1-0) = 2$$

$$\rho_3 = 2(1-1) + 2(1-0) = 2$$

$$\rho_4 = 2(1-0) + 2(1-0.5) + 2(1-0) = 3$$

$$\rho_5 = 2(1-0.5) + 2(1-0) + 2(1-0.5) = 4$$

$$\rho_6 = 2(1-1) + 2(1-0.5) + 2(1-0) = 3$$

$$\Delta x = \Delta y = \frac{1}{2}$$

$$\text{Each particle area} = \Delta x \cdot \Delta y = \frac{1}{4} \Rightarrow r = \frac{1}{4} : 2 = \frac{1}{8}$$

$$W_n = \frac{\pi \hbar^2}{a}$$

Game Physics (IN0037)

Tutorial 6: PDEs and Finite Differences

Exercise 1

For the one-dimensional grid of values below, compute the following quantities:

1. The first derivative at the point x_2 using forward differences.
2. The first derivative at the point x_2 using backward differences.
3. The first derivative at the point x_2 using central differences.
4. The second derivative at the point x_2 using central differences.

The spacing between the points is constant $\Delta x = 0.1$.

$u_i =$	3.2	1.2	-2	-3	0.3	5
	1	2	3	4	5	6
$i =$						

Handwritten notes for Exercise 1:

- First order accuracy
- Forward: $\frac{u_{i+1} - u_i}{\Delta x}$
- Backward: $\frac{u_i - u_{i-1}}{\Delta x}$
- Center: $\frac{u_{i+1} - u_{i-1}}{2\Delta x}$
- Second order accuracy
- Second derivative Central: $\frac{u_{i+1} - 2u_i + u_{i-1}}{(\Delta x)^2}$

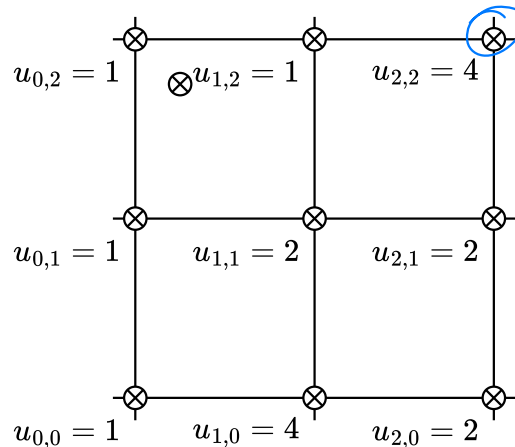
Exercise 2

Rewrite the following symbolic expressions using partial derivative notation (i.e., using $\frac{\partial}{\partial x}$, $\frac{\partial}{\partial y}$, etc.):

1. In 2D: $\nabla \cdot \mathbf{v}$ with $\mathbf{v} = (u \ v)^T$ being a two-dimensional vector field.
2. In 3D: $\Delta \phi = \nabla \cdot \nabla \phi$ with ϕ being a scalar field.
3. In 2D: $\|\nabla \phi\|_2^2$ with ϕ being a scalar field.
4. In 2D: $\nabla \cdot (\mathbf{A} \nabla \phi)$ with $\mathbf{A} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ being a constant 2x2 matrix and ϕ being a scalar field.

Exercise 3

(From Retake 2023/2024) Consider a scalar function in the 2D space $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ which is given on a regular grid with spacing Δx and Δy along each dimension with one value $u_{i,j}$ per node as illustrated below (i being the index for x , j for y). Compute the values of the finite difference expressions below for $\Delta x = \Delta y = 1/2$.



∇u

1. What is the value of $\frac{\partial u}{\partial x}$ for node (2,2) computed with a backward difference?
2. What is the value of $\|\nabla u\|_2^2$ for node (1,1) computed with central differences?
3. What is the value of $\nabla \cdot \nabla u$ for node (1,1) computed with central differences?

2. $\nabla^2 u$

$f(x,y) = 2xy$ - scalar

$f(x,y) = \begin{pmatrix} 2x \\ 5xy^2 \end{pmatrix}$ - vector

gives vector $\rightarrow$

Apply nabla $\rightarrow \nabla f = \begin{pmatrix} \text{part. deriv. } x \\ \text{part. deriv. } y \end{pmatrix} = \begin{pmatrix} 2y \\ 2x \end{pmatrix}$ (gradient)

Apply nabla $\downarrow$

$\nabla \cdot f = \text{der. } x + \text{deriv. } y = 2 + 10xy$ (divergence)

$\nabla^2 f = \nabla \cdot \nabla f$ (Laplace)

tells whether there is a source in a field, 0 $\Rightarrow$ no source

divergence of the gradient

propagation divergence = 0

divergence yes.

$$\|\nabla u\|_2^2 = \left\| \begin{pmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \end{pmatrix} \right\|_2^2 = \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2$$

derivative $\begin{pmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \end{pmatrix}$

Laplacian of scalar func

$$\nabla \cdot \nabla u|_{(1,1)} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

Damping (Energy loss): Затухание

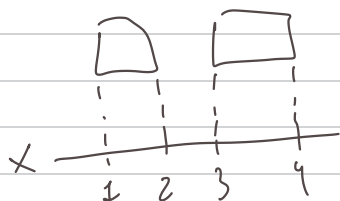
$$F_{\text{damp}}(t) = -c v(t)$$

c - damping coefficient

$v(t)$ - скорость тела.

SAT

Граничное и объект (Separating Axis Test)



← все они ~~находятся~~ на x

$$\left. \begin{array}{l} 2-1=1 \\ 4-3=1 \\ 4-1=3 \end{array} \right\} 2 < 3, \text{ поэтому они не} \\ \text{соприкасаются}$$

Они ~~должны~~ collide на какой-то оси



находятся на x ,
но не на y

sphere box



□ - has to be axis aligned

$$\text{Center} = \frac{1}{4} (\text{сумма точек}) = \frac{1}{4} ((1,1,1) + (1,-1,-1) \dots) = (0,0,0)$$

радиус - расстояние от самой далекой точки до центра